

DEPARTMENT OF ELECTRONICS AND COMPUTER ENGINEERING

Program: B. Tech. (Electronics and Computer Engineering)								Semester: I	
Course: Basic Electronics Engineering								Code: ECES101	
Teaching Scheme (Hrs/week)				Evaluation Scheme (Marks)					
Lecture	Practical	Tutorial	Credit	CIE	ETE	TW	OR	PR	Total
03	02	-	04	40	60	50	-	-	150
Prerequisites:									
1. Basic understanding of electric circuits and components.									
2. Knowledge of basic physics concepts such as electricity, magnetism, and semiconductor behavior.									
3. Understanding of basic digital electronics principles like logic gates and number systems.									
Course Objectives:									
1. To understand the fundamentals of passive electronic components and semiconductor materials.									
2. To master the principles and applications of diodes and special purpose diodes.									
3. To familiarize with transistor operation, configurations, and applications.									
Course Outcomes: After completion of this course, students will able to -									
CO1	Demonstrate proficiency in analyzing and designing electronic circuits utilizing passive components.								
CO2	Explain p-n junction diode and VI characteristics.								
CO3	Apply knowledge of transistor characteristics and configurations in circuit design.								
CO4	Utilize operational amplifiers in electronic circuit design and analysis.								
CO5	Recognize the principles of electronic measurements and instrumentation.								
CO6	Explain basic digital number system conversion.								
Course Contents:									
Unit	Description								Duration (Hrs.)
1.	Introduction to Electronics Components: Introduction to Electronics: Evolution of Electronics, Impact of Electronics in industry and society. Introduction to Passive Components: Classification, Specifications and Color coding techniques of Resistors, Capacitors, Inductors. Introduction to Active Components: Construction, Types and Applications.								6
2.	Semiconductor materials: Semiconductors: P-type and N-type, Current in semiconductors: Diffusion and Drift Current. P-N Junction Diode: Construction, working in forward and reverse bias, V-I characteristics, Diode applications: Diode as a switch, Diode as Rectifier: HWR, FWR, BR, Specifications of Rectifier diodes. Special purpose diodes: Zener diode: V-I Characteristics, Specification and Zener as voltage regulator, Light Emitting Diode (LED) and photo diode.								8
3.	Transistor Circuits: Transistors: Construction, types, operation, Characteristics and region of operation, CB, CE, CC configurations, BJT as a switch and CE amplifier. FET: Introduction, Construction, Operation, V-I characteristics.								8

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	Metal Oxide Semiconductor Field Effect Transistors (MOSFET): Types of MOSFET, n- Channel E-MOSFET : Construction, Operation, V-I characteristics.	
4.	Linear Integrated Circuits: Introduction to Op-amp, Functional block diagram of operational amplifier, ideal practical parameters, Concept of negative & positive feedback, Applications- Inverting and Non inverting amplifier. IC 555 timer as an oscillator, voltage regulation, IC voltage regulators (Three Pin)	6
5.	Electronic Measurements and Instrumentation: Electronics measurements: Frequency measurements and conversions in various units like Hz, KHz, MHz etc, and Voltage, current and power Measurement units, measurement units for resistance, conductance, impedance, capacitance and inductance. Electronic Instruments: Principles and block diagram of digital multimeter, Function Generator, Digital Storage Oscilloscope (DSO) Power scope, AC/DC power supply, Auto transformer, Analog ammeter and voltmeter.	7
6.	Digital Number System and Boolean Algebra: Introduction: Binary, Octal, Decimal, Hexadecimal numbers, and its conversion. Signed Binary number representation: Signed Magnitude, 1's complement and 2's complement representation. Binary, Octal, Hexadecimal Arithmetic: 2's complement arithmetic. Boolean algebra and logic Gates: Boolean algebra, Basic theorems and properties of Boolean algebra. Logic Gates, DeMorgan's theorem.	7
TOTAL		42

List of Experiments:

Perform any Seven (07) experiments from Exp. No 1 to 9, 10th is compulsory:

- Study of Active and Passive components: Resistors (Fixed & Variable), Calculation of resistor value using color code, Capacitors (Fixed & Variable), Inductors, Devices such Diode, BJT, MOSFETs, various IC packages, Switches & Relays.
- Measurements using various measuring equipments:
 - Set up CRO and function generator for measurement of voltage, frequency.
 - Obtain the phase shift between two signals using CRO with the help of Lissagous pattern.
 - Measure voltage, resistance using digital multimeter. Also use multimeter to check diode, BJT.
- Build and test circuits using Semiconductor devices and Plot V-I characteristics:
 - P-N Junction Diode (Study the datasheet of typical PN junction diode 1N 400X).
 - Zener Diode (Study the datasheet of typical Zener diode 1N 4148).
- Build and test Rectifier circuits:
 - Implement half wave, full wave and bridge rectifier using diodes.
 - Observe the effect of capacitor filter on rectifier output.
- Study of Single stage BJT Common Emitter amplifier circuits.
 - Identify pins of a BJT (Such as BC547) and Study its datasheet specifications.
 - To measure voltage and observe waveforms at input and output terminals of single stage BJT Common Emitter amplifier circuits.
 - Calculate Voltage Gain of the amplifier.

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6. Study of Op-amp based amplifier circuits: Build inverting and non-inverting amplifier using op-amp (Study the datasheet of typical Op-Amp 741)
7. Study of IC 555 Timer Circuits.
 - i) Identify pins of IC 555 Timer Circuits.
 - ii) Observe output waveforms and measure frequency of output of IC 555 Timer used in Astable Mode.
8. Study of convergence of number system:
 - i) a) Convert the any number system into its Binary equivalent.
 - ii) b) Convert the any number system into its Octal equivalent.
 - iii) c) Convert the any number system into its Decimal equivalent.
 - iv) d) Convert the any number system into its Hexa decimal equivalent.
9. Verify truth table of Basic Gates.
10. Case Study of any one electronics appliances with block diagram, specification etc.
(Compulsory)

Text Books:

1. Thomas. L. Floyd, "Electronics Devices", 9th Edition, Pearson.
2. R.P. Jain, "Modern Digital Electronics", 4th Edition, Tata McGraw Hill.
3. H.S. Kalsi, "Electronic Instrumentation", 3rd Edition, Tata McGraw Hill.
4. D. Patrnabis, "Sensors and Transducers", 2nd Edition, PHI.

Reference Books:

1. Donald A. Neamen, "Semiconductor Physics and Devices", McGraw-Hill Higher Education, 2011.
2. Paul Horowitz and Winfield Hill, "The Art of Electronics", Cambridge University Press.
3. Ramakant A. Gayakwad, "Op-Amps and Linear Integrated Circuits", Prentice Hall, 2000 Education.
4. Brian R. Jones "Principles of Electronic Instrumentation", Prentice Hall.
5. Ramon Pallas, Areny and John G. Webster, "Sensors and Signal Conditioning", Wiley.

E-Resources:

1. MIT Open CourseWare – Electronics
<https://ocw.mit.edu/courses/6-002-circuits-and-electronics-spring-2007/>
2. NPTEL - Electronics & Communication Engineering
<https://archive.nptel.ac.in/courses/117/105/117105144/>
3. All About Circuits
<https://www.allaboutcircuits.com/>
4. Electronics Hub
<https://www.electronicshub.org/>